सहायक प्राध्यापक (न्यायसहाय्यक विज्ञान) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
Assistant Professor (Forensic Science) Maharashtra Education Service, Group-A
(Collegiate Branch)
Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११६५)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Unit I: Fundamentals of Forensic Science and Criminal Investigations**
	1. Forensic Science Definition, History, Development - Scope, Ethics in Forensic Science
	2. Physical Evidence Nature, Types of Physical Evidence - Search Methods, Collection, Preservation, Packing, and Forwarding of Evidence for Analysis - Chain of Custody
	3. Crime Scene Management Types of Crime Scenes - Preservation and Documentation of Crime Scenes.
	4. Criminal Investigations - Various types of Cases: Unnatural Deaths, Assaults, Sexual Offenses, Poisoning, Vehicular Accidents
	5. Legal Aspects and Court Procedures Types, Powers, and Jurisdiction of Courts - Admissibility of Evidence, Definition of Experts -Relevant Provisions in Bhartiya Nyaya Sanhita (BNS) 2023, Bhartiya Nagarik Suraksha Sanhita (BNSS) 2023, and Bhartiya Sakshya Adhiniyam (BSA) 2023 - Court Procedures, Expert Testimony
	6. Forensic Science Institutions Forensic Labs at Central and State Levels, NFSU, BPRD, NCRB, NCB
	7. Constitutional Rights Right to Equality (Articles 14-18) - Right to Freedom (Articles 19-22)
	8. Criminal Profiling Victim and Culprit Profiling, Role in Investigation Lie Detection (Polygraphy), Narco-Analysis, Brain Mapping
	9. Quality Control in Forensic Labs Management Practices, NABL accreditation
2	**Unit II:Forensic Analytical Techniques**
	1. Microscopy Polarizing, Comparison, Stereoscopic, Fluorescent, Electron Microscopes
	2. Spectrophotometry U V, Visible, IR, Raman, Atomic Absorption, Emission, ATR-FTIR

	3. Neutron Activation Analysis (NAA) Principles, Application in Trace Evidence
	4. X-ray Techniques XRD, XRF Applications, SEM-EDX
	5. Mass Spectrometry Basic Principles, Applications in Forensic Chemistry
	6. Chromatography TLC, GLC, HPLC, HPTLC
	7. Hyphenated Techniques GC-MS, GC-HS, LC-MS, IR-MS, ICP-MS
	8. Electrophoresis Types, Immunoelectrophoresis
	9. Immunoassays Principle, Techniques, Forensic Applications
3	**Unit III: Biological Evidence and DNA Profiling**
	1. Blood Evidence Detection, Species Origin, Blood Group Systems Blood Group Determination Techniques
	2. Body Fluids and Forensic Serology Detection of Seminal and Other Body Fluids, Enzymes, Serum Proteins
	3. Parentage Disputes Paternity and Maternity Testing
	4. DNA Profiling Structure, DNA as a Genetic Marker, DNA Extraction Profiling Techniques, DNA Phenotyping, RNA Profiling
	5. Wildlife Forensics Wildlife Protection Act (1972), Evidence Collection, Identification
4	**Unit IV : Toxicology and Analysis of Poisons**
	1. Alcohol Analysis Ethanol and Methanol Detection, Breath Analysis
	2. Illicit Liquor Analysis, Chemicals in Trap Cases Detection of Chemicals used in Crimes
	3. Toxicology of Poisons Pesticides, OP Compounds, Sedatives, Stimulants, Food poisons
	4. Narcotics and Psychotropic substances and their analysis Opium and opiates , Cannabis & Drugs of abuse their toxicity
	5. Plant (Datura, Castor, Argemone) & Animal poisons-Snakes
	6. Metallic Poisons As, Pb, Th, Zn, Hg their Identification detection and Toxicology
	7. Analytical Techniques for Poison Detection Extraction, Isolation, Viscera Analysis
5	**Unit V: Firearms, Tool Marks and Ballistics**
	1. Firearms and Ammunition Types, Composition, Forensic Examination Velocity, Range and Trajectory GSR analysis
	2. Examination of Country-Made Firearms
	3. Ballistics Internal, External, Terminal and Wound Ballistics.
	4. Tool Marks Types, Erased Mark, Restoration
6	**Unit VI: Fire, Arson, Explosives and Forensic Photography**
	1. Fire and Arson Analysis Petroleum and Incendiary Materials
	2. Explosives Types, Bomb Analysis, IEDs
	3. Arson and Explosion Investigation Protocols & procedures
	4. Forensic Photography Documentation Techniques in Crime Investigation

7	**Unit VII:Trace Evidence, Digital Forensics, Cyberforensics**
	1. Hair and Fiber Analysis Types- Human, animal, Forensic Examination
	2. Pollen, Diatoms Applications in Investigation
	3. Soil and Dust Analysis Composition, Forensic Relevance
	4. Glass, Paint, and Building Materials Forensic Analysis Techniques
	5. Digital Forensics Computer Crimes, Digital Evidence Seizure and Analysis Mobile Forensics
	6. Cybercrimes and Cyberforensics
8	**Unit VIII: Dermatoglyphics/Fingerprints and Biometrics**
	1. Dermatoglyphics/Fingerprint Analysis Classification, Lifting, Chance prints, Examination, Preservation, AFIS
	2. Track Marks Foot, Shoe, Tyre, Skid Marks, Gait Patterns
	3. Biometric Identification Iris scan, Systems and Applications-
	4. Voice Analysis Speaker Identification, Voice Spectrometry, Acoustics, Forensic Relevance
9	**Unit IX: Document and Handwriting Examination**
	1.Document Examination Types, Preliminary Examination Photographic Reproduction Erasures, Alterations, Secret Writings
	2. Ink Analysis, Charred documents analysis
	3.Scientific Examination Techniques
	4. Handwriting and Signature Analysis Identification Techniques Forgeries, Credit Card Examination
10	**Unit X: Forensic Medicine, Anthropology, Odontology and Entomology**
	1. Medicolegal Aspects of Death Types of Death, Post-Mortem examination, Digital autopsy/Virtopsy, various methods of estimating post mortem interval
	2. Injury and Wound Analysis Blunt injuries, Sharp injuries, Gunshot Wounds, Burns
	3. Sexual Offences Current laws pertaining to various sexual offences-rape, sodomy, POCSO Collection of various evidences and their analysis and interpretation
	4. Forensic anthropology and Forensic Odontology Species, Age, Sex, Stature Estimation Anthropometry Skull Superimposition, Bite Marks Role of artificial intelligence
	5. Forensic Entomology Various types of insects used in forensic analysis, their life cycle and role in Estimation of postmortem interval, Entomotoxicology

Note - Wherever Acts/Rule/Regulations are mentioned in syllabus, latest amendments/versions to be considered .

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग